Diseases of the Lower GI System

NURA 340 Pathophysiology | Final Exam Lecture Study Guide

Small intestine, colon, and rectum

Lecture roadmap: Normal intestinal function → Infection, inflammation, and obstruction: *C. diff*, appendicitis, diverticular disease → GI obstruction and ileus → IBS vs. IBD → Crohn's disease → Ulcerative colitis → Crohn's vs. UC → Colorectal cancer (brief) → Lower GI bleeding

How the instructor wants you to think: Connect each disorder to the **normal function it disrupts**, then use that to explain the findings. For example: Why does inflammation cause diarrhea? How can an obstruction lower circulating volume? Why does one IBD cause fistulas while the other causes bloody diarrhea? You don't need every detail. Focus on **the mechanism, the finding it produces, and what happens if it progresses**. Nursing interventions and named exam signs are not tested.

How to read this guide: Everything without a tag comes from the lecture slides and transcript. Anything else is tagged: ADDED information not in the lecture; CLARIFIED a lecture point explained further; CORRECTED an error in the slides or transcript. Every tag is listed in the notes at the end.

Normal Lower GI Function

- **Small intestine**: absorbs **most nutrients** and a large amount of fluid.
- **Colon**: absorbs additional **water and electrolytes** and forms stool.
- **Coordinated muscle contractions** move contents forward.
- The **intestinal lining** is a barrier between bowel contents and the rest of the body.
- The normal **gut microbiome** helps protect against harmful organisms.

Three processes that overlap: **Infection**, **inflammation**, and **obstruction**. When infection or inflammation damages the lining, fluid handling and absorption are impaired (→ diarrhea). When movement slows or something blocks the bowel, contents **accumulate behind the blockage**. Appendicitis, for example, involves obstruction, bacterial growth, inflammation, and eventually impaired blood flow.

Clostridioides difficile Infection (C. diff)

A bacterium that causes inflammation of the colon. It is a leading cause of **healthcare-associated infections** CLARIFIED . It usually causes severe diarrhea and **can be fatal if untreated**.

Etiology	• **Recent antibiotic use** is the biggest risk factor, especially **broad-spectrum**. Antibiotics kill the helpful normal flora that keep other organisms in check. • Higher risk: **older adults**, **weakened immune systems**, **previous C. diff**, **recent healthcare setting**. • Forms **spores** that survive on surfaces; spreads by the **fecal-oral route**.
Pathophysiology	Normal flora disrupted → C. diff multiplies and releases **toxins** → toxins damage cells lining the colon and trigger **inflammation** → the inflamed colon **can't absorb water** → fluid stays in the bowel → **watery diarrhea**. **Colonization vs. infection:** some people **carry** C. diff with no symptoms (colonization). Its presence alone does not mean active infection.
Manifestations	• **Frequent, watery diarrhea** (most common) • Abdominal cramping or tenderness, fever, nausea, decreased appetite • Mucus in stool; blood is less common • A very **distinct odor**
Diagnosis	Symptom pattern + **stool testing**. Concern is raised by new, unexplained diarrhea, especially **≥3 unformed stools in 24 hours**. Tests detect the **toxins** or the organism's **ability to produce them**. Results must be read **with symptoms**, because a colonized person can test positive without active disease.
Impact on health	• **Fluid and electrolyte loss** is the immediate concern → **hypovolemia** → **reduced kidney perfusion** → electrolyte imbalances. The infection is in the colon, but its effects reach far beyond it. • **Pseudomembranes**: collections of inflammatory material over the injured lining • **Toxic megacolon** (severe): the colon dilates and stops working • Severe disease → **perforation, peritonitis, sepsis**

Pharm connection	Antibiotics can cause C. diff, but specific antibiotics also treat it. **Oral vancomycin** is the gold standard. Recurrent disease may need **microbiota-based treatment** (e.g., fecal microbiota transplant) to restore the normal environment. FMT is an interesting fact, not something you're expected to know.

Instructor emphasis: wash your hands. Alcohol-based hand sanitizer does not kill C. diff spores. With any C. diff patient, wash with **soap and water**. The instructor washes for any diarrheal illness, just to be safe.

Appendicitis

Inflammation of the **appendix**, a small pouch attached to the beginning of the large intestine.

Etiology	Often (not always) develops when the **opening of the appendix is blocked**, for example by **hardened stool** or **enlarged lymphoid tissue**.
Pathophysiology	**Blocked opening** → mucus and contents can't drain → **pressure builds** → **bacteria multiply** → **inflammation** → swelling and pressure **compromise blood flow** to the wall → **ischemia** → **necrosis** → **perforation** (contents escape into the abdominal cavity).
Manifestations	• Classic pattern: **vague pain near the umbilicus** that **moves to the right lower quadrant (RLQ)** once inflammation involves the nearby **parietal peritoneum**. The change reflects which tissue is being irritated: organs (viscera) are poor at localizing pain, while the parietal peritoneum localizes it well. • Loss of appetite, nausea, vomiting, fever • Not everyone follows the classic pattern. **No migrating pain does not rule it out**, and RLQ pain alone doesn't rule it in.
Diagnosis	History + abdominal findings + labs + **imaging**. **RLQ tenderness** supports it. **WBC and CRP** may be elevated from inflammation but are **not specific**. **Ultrasound or CT** identifies the inflamed appendix and complications. Know **McBurney's point** (RLQ landmark); you won't be tested on the other named signs (obturator, psoas, Rovsing).

Impact on health	The major concern is progression from **inflammation → tissue death → perforation**. Contained leak → **abscess**; spread through the abdomen → **peritonitis → sepsis** and organ dysfunction. **Temporary pain relief does not mean it has resolved**; tissue injury may still be progressing.
Management	**Appendectomy** is commonly the definitive treatment. Some uncomplicated cases may be treated with **antibiotics** instead (the provider decides).

Diverticular Disease

Diverticula are small **pouches that push out through weak areas** in the colon wall. They develop from changes in the bowel wall plus **pressure inside the colon**. Risk **increases with age**, as tissue integrity declines.

Three terms to keep separate:

- **Diverticulosis** = pouches are **present**. Often no symptoms.
- **Diverticulitis** = **inflammation** in and around a pouch (*-itis*), sometimes with a small perforation.
- **Diverticular bleeding** = a **vessel associated with a pouch** bleeds, **even without diverticulitis**. Usually **painless rectal bleeding**, and should not be extensive.

Manifestations	Diverticulosis: usually none. Diverticulitis: localized abdominal pain and tenderness, often in the **left lower quadrant (LLQ)**; **fever**; changes in bowel habits.
Diagnosis	**CT** is commonly used for suspected acute diverticulitis and complications. **Colonoscopy** can find diverticula but is **usually deferred during acute diverticulitis** because the inflamed tissue risks **perforation**.
Impact on health	**Abscess** or larger **perforation**; repeated inflammation and scarring can **narrow the bowel**; a **fistula** (an abnormal opening between the bowel and another structure) can form.
Management	Depends on severity. **Antibiotics** are often used when infection risk or complications are greater, but they are not always needed in uncomplicated cases.

Discussion Question #1: Why does diverticulitis typically cause LLQ pain?
Diverticula commonly develop in the **sigmoid colon**, which sits in the **left lower abdomen**. Inflammation there causes localized LLQ pain and tenderness. Location links the affected anatomy to the inflammation. However, diverticulitis can occur elsewhere in the colon, and visceral pain can be referred, so this is a common pattern, not a rule.

Intestinal Obstruction and Ileus

Anything that interferes with the **normal forward movement** of intestinal contents. It can be **partial or complete**.

Mechanical obstruction	Functional obstruction (ileus)
A **physical blockage**: **adhesions** after surgery, **hernia**, **tumor**, **stricture**	**No physical blockage**. Coordinated intestinal movement **slows or stops**: after **surgery**, **electrolyte disturbances**, **opioids**
Gas, fluid, and contents collect **above the blockage**; the bowel stretches	Contents accumulate because the **muscle activity** needed to move them is impaired
Bowel contracts against the blockage → **waves of cramping pain**. Bowel sounds **high-pitched early**, then **decreased later**.	Bowel sounds usually **reduced**

Pathophysiology	Fluid becomes **trapped in the bowel** and moves into the bowel wall, so **less stays in the circulation**. If pressure compromises blood flow, especially in a **strangulated** obstruction → **ischemia and tissue death**. Obstruction affects fluid balance and perfusion, not just elimination.
Manifestations	**Abdominal distension**, **nausea and vomiting**, cramping (mechanical), and changes in bowel sounds. **Bowel sounds alone can't make the diagnosis.** A complete obstruction eventually stops **passage of stool and gas**; early on, material already beyond the blockage may still pass.
Diagnosis	**Imaging** shows whether there's a blockage, where, and any complications. **Blood tests** check electrolytes, dehydration, kidney function, and inflammation.

Impact on health	• Vomiting + fluid trapped in the bowel → **reduced circulating volume** → electrolyte disturbances • Vomiting stomach acid → **metabolic alkalosis** (loss of acid) • Compromised blood flow → **ischemia** → **necrosis** → **perforation**
Management	Depends on the cause and whether the bowel is threatened. Some cases resolve with **decompression**: pressure from trapped gas and fluid has to be relieved some other way. **Pharm connection: opioids reduce intestinal motility**, which is why they cause constipation and can contribute to ileus.

Discussion Question #2: After abdominal surgery, a person has distension, nausea, and absent bowel sounds. What mechanism?
Postoperative ileus. Anesthesia, medications, and handling of the bowels during surgery make coordinated movement **slow down**. Gas and fluid accumulate with **no physical blockage**. Always ask: **is movement impaired, or is something physically blocking the bowel?**

IBS vs. IBD: The Big Picture

Both can cause abdominal pain and changes in bowel habits, but the underlying processes are **different**.

- **IBS (irritable bowel syndrome):** a problem with **gut–brain communication**, how the bowel **moves**, and how it **senses** stretching. Symptoms occur **without visible damage** to the lining.
- **IBD (inflammatory bowel disease): inflammation that damages intestinal tissue**. The two main types are **Crohn's disease** and **ulcerative colitis**.

Irritable Bowel Syndrome (IBS)

Etiology	A combination of factors: **gut–brain (nervous system) communication**, **microbiome changes**, a **previous intestinal infection**, and individual responses to **food or stress**.

Pathophysiology	• The bowel may move **too fast** (IBS with diarrhea), **too slow** (IBS with constipation), or **inconsistently** (alternating). • It may be **hypersensitive to stretching**: gas or distension that barely bothers one person can be very painful in IBS. • The gut–brain connection does **not** mean symptoms are imaginary. The nervous system is actively changing sensation and bowel function.
Manifestations	**Recurring abdominal pain related to bowel movements**; change in stool **frequency or form** (diarrhea, constipation, or both); **bloating** is very common.
Diagnosis	Based on the **symptom pattern** (the **Rome criteria**, which you'll see in ATI but don't need to memorize) plus ruling out other causes. **Bleeding, unexplained weight loss, or anemia** need further evaluation, because **IBS does not explain them**.
Impact on health	No destructive tissue injury, but symptoms can be **substantial**: they affect work, school, eating, and social life. People may plan their day around discomfort or bathroom access.
Management	**Individualized** to the pattern: medications may target constipation, diarrhea, pain, or gut sensitivity.

Crohn's Disease

A **chronic, immune-mediated** form of IBD. It develops from **genetic susceptibility**, **environmental triggers**, the **microbiome**, and an **abnormal immune response** in the GI tract. The exact cause is unknown.

Two features to know (instructor emphasis):

1. **Patchy distribution, called "skip lesions."** It can affect **any part of the GI tract** (mouth to anus), most often the **terminal ileum and colon**. Inflamed areas are separated by relatively normal tissue.
2. **Depth: transmural.** Inflammation can go through the **full thickness** of the bowel wall.

Pathophysiology	• Fewer **Paneth cells**, and the ones present are often mutated. The lining becomes infiltrated with inflammatory cells → **hyperemia and edema** of the mucosa. (You don't need cell-level details.) • **Ulcerations** form over the inflamed areas (the skip lesions). • Chronic inflammation → **thickened bowel wall**. Repeated inflammation and healing → **narrowing (strictures)**. • Because inflammation is **transmural**, Crohn's can cause **abscesses and fistulas**.
Manifestations	Depend on location and extent: **diarrhea**, **abdominal cramping**, **fatigue**, **weight loss**. Blood may be present, but not everyone has bloody diarrhea. **Small-intestine** involvement → impaired absorption → **nutritional deficiencies**; reduced intake adds to weight loss. Inflammation **outside the GI tract**: **joints, eyes, skin** (common with immune-mediated disease).
Diagnosis	**Blood tests** (anemia, inflammation); **stool tests** (intestinal inflammation, rule out infection); **colonoscopy with biopsy** and **imaging** show location, pattern, and extent.
Impact on health	Complications follow the **depth** of inflammation: • **Stricture** → narrowed passage → **obstruction** • **Fistula** → abnormal connection between the bowel and another structure • **Abscess** → localized collection of infection • Ongoing inflammation, diarrhea, and poor absorption affect **hydration, nutrition, bone health, and electrolytes**.
Pharm connection	Goals: control inflammation, achieve **remission**, prevent further damage. **Corticosteroids** for **flares** (**not long-term**); **immune-modifying and biologic** drugs for maintenance.

Ulcerative Colitis (UC)

Another **chronic, immune-mediated** IBD from interacting genetic, immune, environmental, and microbiome factors. It differs from Crohn's in **location, distribution, and depth**.

Location and pathophysiology	• Affects only the **colon and rectum**. • Usually **starts in the rectum** and extends **upward in a continuous pattern** (no skip lesions). • Involves mainly the **inner lining (mucosa)**, not the full wall. The surface becomes **inflamed, fragile, and ulcerated**.
Manifestations	• **Bloody diarrhea with mucus** (key symptom): inflamed surface tissue **bleeds easily**, and impaired fluid absorption makes stools looser. • **Urgency** and **tenesmus** (a persistent feeling of needing to pass stool even when little is left) from rectal inflammation • Abdominal cramping, fatigue • Occurs in **flares** with periods of improvement or **remission**
Diagnosis	**Blood and stool tests** (inflammation, anemia, infection); **endoscopy with biopsy** shows the characteristic pattern.
Impact on health	Repeated diarrhea → **dehydration and electrolyte loss**; bleeding → **anemia**; severe inflammation → **toxic megacolon**, significant **hemorrhage**, **perforation**. **Long-standing UC increases colon cancer risk.**
Pharm connection	Reduce inflammation and maintain remission: **corticosteroids** for flares; **immune-modifying/biologic** therapy. Controlling inflammation lets the lining heal.

Crohn's Disease vs. Ulcerative Colitis

Focus on **location, distribution, and depth**. Those are the keys to telling them apart.

Feature	Crohn's disease	Ulcerative colitis
Location	**Any part** of the GI tract; often terminal ileum and colon	**Colon and rectum** only
Distribution	**Patchy**: skip lesions	**Continuous**, usually starting in the rectum
Depth	**Full thickness** of the bowel wall (transmural)	Mainly the **inner lining** (mucosa)

Feature	Crohn's disease	Ulcerative colitis
Common findings	Abdominal pain, diarrhea, fatigue, weight loss; bleeding may occur	**Bloody diarrhea**, urgency, cramping, fatigue
Complications	**Strictures, fistulas, abscesses**, nutritional deficiencies	Anemia, dehydration, **toxic megacolon**, perforation; ↑ colon cancer risk
Shared	Chronic inflammation; **flares and remission**; pain, diarrhea, fatigue, nutritional problems. Complications can overlap.	

Memory aid ADDED **Crohn's = "Crawls everywhere, Cuts deep."** It can appear anywhere from mouth to anus, skips around, and goes through the whole wall (fistulas). **UC = "Up from the rectum, Continuous, Upper layer only."** It starts at the rectum, spreads up without gaps, and stays in the mucosa (bloody diarrhea).

IBS vs. IBD

Feature	IBS	IBD (Crohn's and UC)
Primary problem	Altered gut–brain interaction, motility, and sensitivity	Immune-mediated inflammation
Tissue changes	**No** destructive inflammation or ulceration	Inflammation **damages** the lining or bowel wall
Manifestations	Abdominal pain, bloating, diarrhea, constipation, or both	Diarrhea, pain, fatigue; may include bleeding and weight loss
Bleeding	**Not explained by IBS**; needs further evaluation	May occur from inflamed or ulcerated tissue
Nutrition	Absorption generally preserved; food restrictions may affect intake	Reduced intake, nutrient loss, and poor absorption → deficiencies
Complications	Can substantially affect quality of life	Anemia, malnutrition, bowel damage
Pharm connection	Target bowel movement patterns and discomfort	Reduce inflammation, modify the immune response

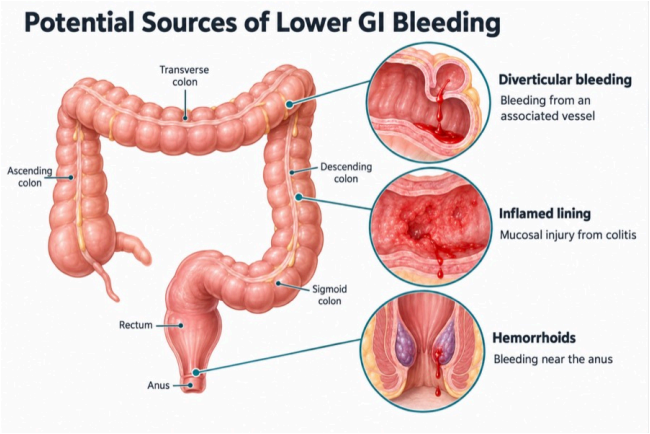
Pain and diarrhea occur in both, so **symptoms alone can't tell them apart**. Start with what is happening **in the tissue**.

Colorectal Cancer (Brief Connection)

- Covered in detail later in the **hematology and cancer module**. The instructor said **not to review the ATI colorectal cancer content yet**.
- Link here: some IBD (especially long-standing UC) **increases colon cancer risk**.
- Still more common in older adults, but rates have been **rising in adults under 50**.
- **Two findings that need further evaluation: unexplained blood in the stool** and a **persistent change in usual bowel habits** (a real change, not once or twice). These don't automatically mean cancer, but **don't dismiss them** as hemorrhoids or IBS, especially just because the person is young.

Lower GI Bleeding

Usually refers to bleeding from the **colon or rectum**. Bleeding is a **finding that needs an explanation**, not a single disease.



From the lecture slides: diverticular bleeding, inflamed lining (colitis), and hemorrhoids.

Causes and mechanisms	• **Diverticular bleeding**: a vessel associated with a pouch is damaged • **Inflammatory or infectious colitis**: inflammation or ulceration injures the mucosal surface • **Abnormal blood vessels**, e.g., **hemorrhoids** • **Tumors**: vessels damaged as the tumor grows

Manifestations	Blood may be **bright red or maroon**, mixed with stool or separate. **Frank** bleeding = visible. **Occult** bleeding = not visible, found only by testing. Remember: a **rapid upper GI bleed** can also produce red blood from the rectum, so appearance guides evaluation but can't establish the source.
Diagnosis	Considers the bleeding pattern, associated symptoms, and effects on circulation. **Blood tests** check for anemia; **colonoscopy or imaging** finds the source.
Impact on health	**Slow** bleeding → **iron deficiency anemia**. **Rapid or large** bleeding → **reduced circulating volume** and compromised perfusion. Whether upper or lower, the systemic effect depends on **how much and how fast**.
Management	Stop the bleeding **and** treat the effects of blood loss. **Pharm connection: anticoagulants can make bleeding worse**, but they don't explain where the injury is, so the source still needs to be found.

Quick Review

Disorder	Core mechanism	Hallmark findings	Key complications
C. diff	Antibiotics disrupt flora → toxins inflame the colon → can't absorb water	Frequent watery diarrhea (≥3 unformed/24 h), cramping, fever, distinct odor	Hypovolemia, AKI, pseudomembranes, toxic megacolon, sepsis
Appendicitis	Blocked opening → pressure → bacteria → inflammation → ischemia	Periumbilical pain → RLQ; anorexia, N/V, fever; ↑WBC/CRP	Necrosis, perforation, abscess, peritonitis, sepsis
Diverticular disease	Pouches through weak spots in the colon wall (-osis); inflamed (-itis)	Diverticulitis: LLQ pain, fever; bleeding is painless	Abscess, perforation, narrowing, fistula
Obstruction / ileus	Physical blockage vs. impaired motility	Distension, N/V, cramping; high-pitched → absent bowel sounds	Hypovolemia, metabolic alkalosis, ischemia, perforation

Disorder	Core mechanism	Hallmark findings	Key complications
IBS	Gut–brain, motility, and sensitivity problem; no tissue damage	Pain with BMs, diarrhea and/or constipation, bloating	Quality of life (bleeding/weight loss = not IBS)
Crohn's	Immune-mediated, patchy, transmural, anywhere in the GI tract	Diarrhea, cramping, weight loss, malabsorption	Strictures, fistulas, abscesses
UC	Immune-mediated, continuous from the rectum, mucosa only	Bloody diarrhea with mucus, urgency, tenesmus	Anemia, toxic megacolon, perforation, colon cancer
Lower GI bleed	Diverticular vessel, colitis, hemorrhoids, tumor	Bright red/maroon blood; frank or occult	Iron deficiency anemia (slow); shock (rapid)

Instructor wrap-up: Ask what is interfering with normal intestinal function: Is infection damaging the lining? Is an abnormal immune response causing ongoing inflammation? Are contents building up because movement slowed or something is blocking the bowel? Then connect the mechanism to the findings. **Inflammation impairs absorption and causes diarrhea; deep injury causes fistulas or perforation; obstruction traps fluid and threatens blood flow.**

Self-Check ADDED

1. Why is recent antibiotic use the biggest risk factor for C. diff?
   Antibiotics kill the normal flora that keep C. diff from overgrowing.
2. Why must you use soap and water instead of hand sanitizer with C. diff?
   Alcohol doesn't kill C. diff spores.
3. A patient tests positive for C. diff but has formed stools and no symptoms. What does that mean?
   Colonization, not active infection.
4. Why does appendicitis pain move from the umbilicus to the RLQ?
   Early pain is visceral and poorly localized. Once inflammation reaches the parietal peritoneum, pain localizes to the RLQ.

5. A patient with appendicitis suddenly says the pain is better. Is this reassuring?
   No. Temporary relief can occur while tissue injury progresses, including after perforation.
6. Diverticulosis vs. diverticulitis?
   -osis: pouches are present, usually no symptoms. -itis: the pouches are inflamed, with LLQ pain and fever.
7. Why is colonoscopy usually avoided during acute diverticulitis?
   Risk of perforating the inflamed tissue.
8. How can a bowel obstruction cause hypovolemia?
   Fluid is trapped in the bowel and bowel wall, and vomiting adds to the loss, so less stays in the circulation.
9. Why can vomiting from an obstruction cause metabolic alkalosis?
   Loss of stomach acid (HCl).
10. Which IBD is more likely to cause a fistula, and why?
    Crohn's, because inflammation is transmural (through the whole wall).
11. A patient has bloody diarrhea with mucus, urgency, and tenesmus. Inflammation starts at the rectum and is continuous. Which IBD?
    Ulcerative colitis.
12. A patient with presumed IBS develops rectal bleeding and weight loss. What do you do?
    Flag it for further evaluation. IBS does not explain bleeding, weight loss, or anemia.
13. Frank vs. occult bleeding?
    Frank is visible. Occult is not visible and is found only by testing.

Notes on Sources, Additions & Clarifications

This guide was written from the lecture slides (Diseases of the Lower GI System, NURA 340), the comparison tables on the slides, and the organized lecture transcript. Most student slides contain only headings and images, so most content comes from the transcript.

- CORRECTED **C. diff name (slide 4).** The slide title says "Clostridium difficile" with the note "Used to be called Clostridioides." This is reversed. The **current** name is *Clostridioides difficile*; it **used to be** called *Clostridium difficile*. The abbreviation "C. diff" is the same either way.
- CLARIFIED **"Most common cause of healthcare-associated infections."** The lecturer says C. diff is *the* most common cause of HAIs. It is one of the most common, and is often cited as the most common cause of healthcare-associated **infectious diarrhea**. The guide says "a leading cause" to stay accurate. For the exam, remember it is a major HAI.
- CLARIFIED **Visceral vs. parietal pain (appendicitis).** The lecturer says organs are poor at localizing pain; I added that the parietal peritoneum localizes pain well, which explains the shift to the RLQ.
- CLARIFIED **Named appendicitis signs.** The lecturer said not to memorize the obturator, psoas, and Rovsing signs but to "know McBurney's point." They are named here only so you can recognize them.
- ADDED **Crohn's vs. UC memory aid** is mine, not from the lecture.
- ADDED **Self-Check questions** are my own, written from lecture content.
- Colorectal cancer is intentionally brief. The instructor said it will be covered in the hematology and cancer module.